

Program:	B. Tech. (Computer Engineering)				Semester: VI			
Course:	Cloud Computing				Code: BCE26PE07			
Credit	Teaching Scheme (Hrs./Week)				Evaluation Scheme and Marks			
	Lecture	Practical	Tutorial	Other	FA		SA	Total
					FA1	FA2		
2	2	-	-	1	10	10	30	50
Prior knowledge of Computer Networks ,Operating Systems and Database Management System is essential.								
Course Objectives: This course aims at enabling students: 1. To introduce various cloud computing services and models. 2. To learn how to select the appropriate configuration to compute a node. 3. To configure appropriate storage service for the specific application. 4. To understand networking and security settings in a cloud environment. 5. To design the application using high scalability and reliability considerations.								
Course Outcomes: After learning the course, the students will be able to: 1. Understand various cloud computing services, models and virtualization. 2. Analyze the various storage-related services while computing.. 3. Apply appropriate database services during application deployment. . 4. Apply various networking and security options to the cloud..								
Detailed Syllabus								
Unit	Description						Duration (Hrs)	
I	Introduction to Cloud Computing: Introduction, Characteristics of Cloud Computing, Cloud Models, Cloud Services Examples, Cloud-based Services and Applications, Cloud Concepts and Technologies. Virtualization: Using Virtualization Technologies, Load Balancing and Virtualization, Understanding Hypervisors, VMware vSphere, Understanding Machine Imaging, Capacity Planning. Basics of Containerization with Docker and Kubernetes.						8	
II	Web Services: Understanding Amazon Web Services, Amazon Web Service Components and Services, working with the Elastic Compute Cloud (EC2), Working with Amazon Storage Systems, Exploring Platform as a Service: Defining Services, SaaS versus PaaS.						7	
III	Database Services in Cloud Computing: Need for cloud databases, considerations for cloud databases, architecture and common characteristics, Data Models Relational database (MySQL), Key-value-based database (DynamoDB), In-memory database, Document based database (Mongo DB), Graph based database (Neo4j), Time series database, Ledger based database.						7	

IV	Security in Cloud Computing: Risks in Cloud Computing: Risk Management, Enterprise-Wide Risk Management, Types of Risks in Cloud Computing. Data Security in Cloud: Security Issues, Challenges, Advantages, Disadvantages, Cloud Digital persona and Data security. Cloud Security Services: Confidentiality, Integrity and Availability, Security Authorization Challenges in the Cloud, Secure Cloud Software Requirements, Secure Cloud Software Testing. High Availability and Scalability Services in Cloud Computing	8
	Self-Study virtual machines and containers, Storage for Backups and databases- in Azure and GCP, Databases, Storage gateway for hybrid cloud storage systems, Time series database, Ledger based database, Cloud Security in Azure and GCP, research paper reading.	6*
Total (*Not Included)		30
Text Books: 1. A.Srinivasan, J.Suresh, "Cloud Computing: A Practical Approach for Learning and Implementation", Pearson, ISBN: 978-81-317-7651-3. 2. Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi Mastering Cloud. Computing McGraw Hill Education, 2023. 3. Wittig, Michael, Andreas Wittig, and Ben Whaley. Amazon web services in action. Manning, 2022. 4. Murty, James. Programming amazon web services: S3, EC2, SQS, FPS, and SimpleDB, "O'Reilly Media, Inc.", 2015.		
Reference Books: 1. Kavis, Michael J. Architecting the cloud: design decisions for cloud computing service models (SaaS, PaaS, and IaaS). John Wiley Sons, 2024. 2. Vacca, John R., ed. Cloud computing security: foundations and challenges. CRC Press, 2023. 3. Furht, Borivoje, and Arman do Escalante. Handbook of cloud computing. Vol. 3. New York: Springer, 2021. 4. Piper, Ben, and David Clinton. AWS Certified Solutions Architect Study Guide: Associate SAA-C02 Exam. John Wiley Sons, 2020. 5. Anthony, Albert. Mastering AWS Security: Create and maintain a secure cloud ecosystem. Packt Publishing Ltd, 2017.		
Online Sources: 1. Cloud Computing by Soumya Ghosh, https://nptel.ac.in/courses/106105167 2. Indian Institute of Technology Delhi: https://library.iitd.ac.in/arpit_2020-2021/Week%2016%20-%20Module%2039%20-%20PPT-%20Cloud%20Computing%20and%20Its%20Application%20in%20Libraries.pdf 3. IIT Roorkee: Intellipaat.com/advanced-certification-cloud-computing. 4. Cloud computing By Prof. Soumya Kanti Ghosh IIT Kharagpur: https://onlinecourses.nptel.ac.in/noc21_cs14/preview.		